A Periodic Table of the Finite Simple Groups

Every finite simple group appears exactly once, in its simplest form. Columns are grouped by rank; the vertical axis is the field size q.

TYPE

- 🟥 Cyclic groups
- 🟪 Alternating groups
- 🟦 Chevalley groups
- 🟩 Steinberg groups
- 🟨 Suzuki and Ree groups
- 🟪 Sporadic groups

HOW TO READ A CELL

≅ $A_2(2)$ — other descriptions of the same group

$A_1(7)$ — the group

$2^3 \cdot 3 \cdot 7$ — prime factorisation of its order

AN EMPTY CELL

the group is not defined there, or is not simple,
or it is isomorphic to one printed elsewhere —
every group appears only under its simplest name:

$A_2(1) \to C_3$
$A_1(2^2), A_1(5) \to A_5\ [A_4(1)]$
$A_1(3^2) \to A_6\ [A_5(1)]$

$A_2(2) \to A_1(7)$
$A_3(2) \to A_8\ [A_7(1)]$
${}^2A_3(2) \to B_2(3)$

$C_n(q)$, q even $\to B_n(q)$
$B_2(2)' \to A_6\ [A_5(1)]$
$G_2(2)' \to {}^2A_2(3)$

${}^2G_2(3)' \to A_1(2^3)$

FAMILIES

$A_n(q) = L_{n+1}(q)$ — Linear
$B_n(q) = O_{2n+1}(q)$ — Orthogonal (odd)
$C_n(q) = S_{2n}(q)$ — Symplectic
$D_n(q) = O^+_{2n}(q)$ — Orthogonal (plus)

$G_2(q)$ — Exceptional G_2
$F_4(q)$ — Exceptional F_4
$E_6(q)$ — Exceptional E_6
$E_7(q)$ — Exceptional E_7

$E_8(q)$ — Exceptional E_8
${}^2A_n(q) = U_{n+1}(q)$ — Unitary
${}^2D_n(q) = O^-_{2n}(q)$ — Orthogonal (minus)
${}^3D_4(q)$ — Triality

${}^2E_6(q)$ — Twisted E_6
${}^2B_2(q)$ — Suzuki
${}^2G_2(q)$ — Ree (small)
${}^2F_4(q)$ — Ree (large)

cyclic
rank 1
rank 2
rank 3
rank 4
rank 5
rank 6
rank 7
rank 8
rank 9
rank n

C_p (p prime)	p
$A_1(q)$	$q(q^2-1)/\gcd(2,q-1)$
$A_2(q)$	$q^3(q^3-1)(q^2-1)/\gcd(3,q-1)$
$B_2(q)$	$q^4(q^4-1)(q^2-1)/\gcd(2,q-1)$
$G_2(q)$	$q^6(q^6-1)(q^2-1)$
${}^2A_2(q)$	$q^3(q^3+1)(q^2-1)/\gcd(3,q+1)$
${}^2B_2(2^{2m+1})$	$2^{4m+2}(2^{4m+2}+1)(2^{2m+1}-1)$
${}^2G_2(3^{2m+1})$	$3^{6m+3}(3^{6m+3}+1)(3^{2m+1}-1)$
$A_3(q)$	$q^6(q^4-1)(q^3-1)(q^2-1)/\gcd(4,q-1)$
$B_3(q)$	$q^9(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_3(q)$	$q^9(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
${}^2A_3(q)$	$q^6(q^4-1)(q^3+1)(q^2-1)/\gcd(4,q+1)$
$A_4(q)$	$q^{10}(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(5,q-1)$
$B_4(q)$	$q^{16}(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_4(q)$	$q^{16}(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$D_4(q)$	$q^{12}(q^6-1)(q^4-1)(q^4-1)(q^2-1)/\gcd(4,q^4-1)$
$F_4(q)$	$q^{24}(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)$
${}^2A_4(q)$	$q^{10}(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(5,q+1)$
${}^2D_4(q)$	$q^{12}(q^6-1)(q^4+1)(q^4-1)(q^2-1)/\gcd(4,q^4+1)$
${}^3D_4(q)$	$q^{12}(q^8+q^4+1)(q^6-1)(q^2-1)$
${}^2F_4(2^{2m+1})$	$2^{24m+12}(2^{12m+6}+1)(2^{8m+4}-1)(2^{6m+3}+1)(2^{2m+1}-1)$
$A_5(q)$	$q^{15}(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(6,q-1)$
$B_5(q)$	$q^{25}(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_5(q)$	$q^{25}(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$D_5(q)$	$q^{20}(q^8-1)(q^6-1)(q^5-1)(q^4-1)(q^2-1)/\gcd(4,q^5-1)$
${}^2A_5(q)$	$q^{15}(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(6,q+1)$
${}^2D_5(q)$	$q^{20}(q^8-1)(q^6-1)(q^5+1)(q^4-1)(q^2-1)/\gcd(4,q^5+1)$
$A_6(q)$	$q^{21}(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(7,q-1)$
$B_6(q)$	$q^{36}(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_6(q)$	$q^{36}(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$D_6(q)$	$q^{30}(q^{10}-1)(q^8-1)(q^6-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^6-1)$
$E_6(q)$	$q^{36}(q^{12}-1)(q^9-1)(q^8-1)(q^6-1)(q^5-1)(q^2-1)/\gcd(3,q-1)$
${}^2A_6(q)$	$q^{21}(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(7,q+1)$
${}^2D_6(q)$	$q^{30}(q^{10}-1)(q^8-1)(q^6+1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^6+1)$
${}^2E_6(q)$	$q^{36}(q^{12}-1)(q^9+1)(q^8-1)(q^6-1)(q^5+1)(q^2-1)/\gcd(3,q+1)$
$A_7(q)$	$q^{28}(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(8,q-1)$
$B_7(q)$	$q^{49}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_7(q)$	$q^{49}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$D_7(q)$	$q^{42}(q^{12}-1)(q^{10}-1)(q^8-1)(q^7-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^7-1)$
$E_7(q)$	$q^{63}(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^2-1)/\gcd(2,q-1)$
${}^2A_7(q)$	$q^{28}(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(8,q+1)$
${}^2D_7(q)$	$q^{42}(q^{12}-1)(q^{10}-1)(q^8-1)(q^7+1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^7+1)$
$A_8(q)$	$q^{36}(q^9-1)(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(9,q-1)$
$B_8(q)$	$q^{64}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_8(q)$	$q^{64}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$D_8(q)$	$q^{56}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^8-1)$
$E_8(q)$	$q^{120}(q^{30}-1)(q^{24}-1)(q^{20}-1)(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^2-1)$
${}^2A_8(q)$	$q^{36}(q^9+1)(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(9,q+1)$
${}^2D_8(q)$	$q^{56}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8+1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^8+1)$
$A_9(q)$	$q^{45}(q^{10}-1)(q^9-1)(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(10,q-1)$
$B_9(q)$	$q^{81}(q^{18}-1)(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$C_9(q)$	$q^{81}(q^{18}-1)(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
$D_9(q)$	$q^{72}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^9-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^9-1)$
${}^2A_9(q)$	$q^{45}(q^{10}-1)(q^9+1)(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(10,q+1)$
${}^2D_9(q)$	$q^{72}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^9+1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^9+1)$
$A_n(q)$	$q^{n(n+1)/2}\prod_{i=2}^{n+1}(q^i-1)/\gcd(n+1,q-1)$
$B_n(q)$	$q^{n^2}\prod_{i=1}^{n}(q^{2i}-1)/\gcd(2,q-1)$
$C_n(q)$	$q^{n^2}\prod_{i=1}^{n}(q^{2i}-1)/\gcd(2,q-1)$
$D_n(q)$	$q^{n(n-1)}(q^n-1)\prod_{i=1}^{n-1}(q^{2i}-1)/\gcd(4,q^n-1)$
${}^2A_n(q)$	$q^{n(n+1)/2}\prod_{i=2}^{n+1}(q^i-(-1)^i)/\gcd(n+1,q+1)$
${}^2D_n(q)$	$q^{n(n-1)}(q^n+1)\prod_{i=1}^{n-1}(q^{2i}-1)/\gcd(4,q^n+1)$

Sporadic groups

Mathieu	M_{11}	$2^4 \cdot 3^2 \cdot 5 \cdot 11$
Mathieu	M_{12}	$2^6 \cdot 3^3 \cdot 5 \cdot 11$
Mathieu	M_{22}	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$
Mathieu	M_{23}	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$
Mathieu	M_{24}	$2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$
Janko	J_1	$2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$
Hall–Janko	J_2	$2^7 \cdot 3^3 \cdot 5^2 \cdot 7$
Janko	J_3	$2^7 \cdot 3^5 \cdot 5 \cdot 17 \cdot 19$
Janko	J_4	$2^{21} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11^3 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 43$
Higman–Sims	HS	$2^9 \cdot 3^2 \cdot 5^3 \cdot 7 \cdot 11$
McLaughlin	McL	$2^7 \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11$
Held	He	$2^{10} \cdot 3^3 \cdot 5^2 \cdot 7^3 \cdot 17$
Rudvalis	Ru	$2^{14} \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13 \cdot 29$
Suzuki	Suz	$2^{13} \cdot 3^7 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$
O'Nan	O'N	$2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31$
Conway	Co_1	$2^{21} \cdot 3^9 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$
Conway	Co_2	$2^{18} \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$
Conway	Co_3	$2^{10} \cdot 3^7 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$
Fischer	Fi_{22}	$2^{17} \cdot 3^9 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$
Fischer	Fi_{23}	$2^{18} \cdot 3^{13} \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23$
Fischer	Fi_{24}'	$2^{21} \cdot 3^{16} \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29$
Harada–Norton	HN	$2^{14} \cdot 3^6 \cdot 5^6 \cdot 7 \cdot 11 \cdot 19$
Lyons	Ly	$2^8 \cdot 3^7 \cdot 5^6 \cdot 7 \cdot 11 \cdot 31 \cdot 37 \cdot 67$
Thompson	Th	$2^{15} \cdot 3^{10} \cdot 5^3 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31$
Baby Monster	B	$2^{41} \cdot 3^{13} \cdot 5^6 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47$
Monster	M	$2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$

CLASSIFICATION THEOREM

Every finite simple group is cyclic of prime order, alternating of degree n ≥ 5, a group of Lie type, or one of the 26 sporadic groups. This poster shows every one of them, each exactly once.

NOTES

- After the cyclic column the columns are grouped by rank: every family with a simple group of rank 1, then of rank 2, and so on.
- E_8 is the last family to appear, so from rank 9 onwards every block is the same six classical families — hence the ellipsis.
- Row q = 1: over "the field with one element" a group of Lie type degenerates to its Weyl group — for type A_n the symmetric group S_{n+1} — whose simple part is A_{n+1}, just as the simple part of $GL_{n+1}(q)$ is $A_n(q) = L_{n+1}(q)$. Only type A is simple there, so that row is exactly the alternating groups.
- "$A_n(1)$" is a position on this table, not a received notation: there is no field with one element, and the q → 1 limit is a heuristic of Tits'. Each group in that row is printed under its own name, A_{n+1}.
- A cell carrying a parameter shows the generic order formula in place of a prime factorisation.
- $C_n(q) \cong B_n(q)$ for q even, so only the odd-q cells of the C columns are printed.
- ${}^2B_2(q)$ and ${}^2F_4(q)$ exist for $q = 2^{2m+1}$ and ${}^2G_2(q)$ for $q = 3^{2m+1}$ (m ≥ 1); their other rows are empty.
- ${}^2F_4(2)$ is not simple, but its derived subgroup — the Tits group — is, and is printed in that cell.
- The 26 sporadic groups belong to no family of Lie type; they are shown in their own strip.
- Above each column is the Dynkin diagram of its family. On the Steinberg, Suzuki and Ree columns the red arrows are the automorphism of the diagram that the Frobenius twist uses — for ${}^3D_4$ a turn of order three about the branch node.
- A symbol with a subscript and no field is a group over "F_1": C_p is cyclic of order p and A_m is alternating of degree m. With a field, $A_n(q)$ is linear and $C_n(q)$ symplectic — the argument tells them apart.